Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

AUXILIARY EXAMINATION

B. E. (Fourth Year) | Course Code: UCS-802
Date: 30.08.26 | Course Name: Compiler Design
Time: 3 Hours, M. Marks: 100 | Name of Faculty: Rupali Bhardwaj

Note: Attempt any ten questions. Assume missing data, if any, suitably

Q1	Explain the different phases of compiler in brief. Write down the output of lexical and syntax analyzer for the expression a = b + c * 60	10
Q2	Consider regular expression $a(a+b)^*bb$ and do the following- i. Draw a NFA using Thompson's Construction method. ii. Draw a DFA using subset construction method.	10
Q3	Consider the following Context-Free Grammar along with Semantic actions for each production in $G = \{E, \{E, T, F\}, \{+, \times, id, (,)\}, P\}$, where P is set of productions as: $E \rightarrow E_1 \times T$ $\{E.val = E_1.val * T.val\}$ $E \rightarrow T$ $\{E.val = T.val\}$ $T \rightarrow T_1 + F$ $\{T.val = T_1.val + F.val\}$ $T \rightarrow F$ $\{T.val = F.val\}$ $F \rightarrow id$ $\{F.val = id.lexval\}$ $F \rightarrow (E)$ $\{F.val = E.val\}$ a) Remove left recursion from the above SDT. b) Construct FIRST and FOLLOW sets. c) Construct LL(1) parser for the above grammar. d) Construct the annotated parse tree for expression $3 \times 4 + 5 \times 8$	10
Q4	Consider the following Context-Free Grammar $G = \{\{E, T, F\}, E, \{*, -, +, /, (,), \boldsymbol{id}\}, P\}$ where P is set of productions having elements: $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid \boldsymbol{id}$ a) Construct the LR(0) canonical set for the above grammar. b) Construct $SLR(1)$ parsing table for the above grammar. c) Show the actions of the $SLR(1)$ parser to recognize input string $\boldsymbol{id} * \boldsymbol{id} + \boldsymbol{id}$	10
Q5	Consider the following grammar representing simplified integer expressions in LISP-like prefix form: $NonTerminal: \{lex, atom, list, lexseq\}$ $Terminal: \{number, id, (,)\}$ and productions as: $lex \rightarrow atom \mid list$ $atom \rightarrow number \mid id$ $list \rightarrow (lexseq)$ $lexseq \rightarrow lexseq\ lex \mid lex$ Find a rightmost derivation for (*a 23 (m x y)*)	10 PTO

Q6	Describe various error recovery strategies that can be adopted at various stages of compiler design.	10
Q7	a) Explain in brief the different ways of passing the parameters to the procedure with help of some examples. b) Indicate the handles in the reduction of right senetial form $SS + a *$ to the start symbol using grammar $S \rightarrow SS + \mid SS *\mid a$	10
Q8	Consider the flow diagram given below. $x = 3;$ $y = 2;$ $L1:$ $a = 5.2;$ $b = 9.4;$ $\boldsymbol{if}\ (a > 5)$ $k = x * y;$ $c = 3.8;$ $\boldsymbol{else}$ $m = x * y;$ $a = y + b;$ $k = x * y;$ $d = c * 2;$ $goto\ L1;$ Can we apply any of the following optimization technique on it? Justify your answer with proper reasoning: i) Common Subexpression Elimination ii) Dead code elimination iii) Constant Propagation iv) Frequency reduction	10
Q9	Given brief reasons (in favor or against) for each of the following statements a) The size of SLR(1) table and LALR(1) table for the same grammar are identical. b) While parsing a sentence of a language, the behavior of LR(1) parser and LALR(1) parser are identical.	10
Q10	With the help of parse trees, determine whether the following grammar is ambiguous or unambiguous for the string: NOT (TRUE OR FALSE AND (NOT TRUE)) The grammar is given as: *bexpr → bexpr OR bterm | bterm* *bterm → bterm AND bfactor | bfactor* *bfactor → NOT bfactor | (bexpr) | TRUE | FALSE* where bexpr is start symbol of grammar, { OR, AND, NOT, TRUE, FALSE, (,)} are terminals whereas {bexpr, bterm, bfactor} are nonterminals .	1
Q11	a) Using suitable examples, differentiate between i. Synthesized and Inherited attributes ii. Dependency graph and Annotated Parse tree b) Explain different representations of 3-address code with the help of a suitable example	10